Gamma-ray Analysis for Multi-Messenger Astrophysics: GAMMA

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I. INTRODUCTION

The Gamma-ray Analysis for Multi-Messenger Astrophysics (GAMMA) initiative is a project by the CAPIBARA Collaboration with the goal to study the high-energy, gamma-ray phenomena in the universe and launch a CubeSat to observe these events. In this white paper, we provide a set of compelling open problems that motivate the development and launch of such a mission.

In the 2020 decadal survey [1], it was pointed out that multi-band, that means acros the electromagnetic spectrum, and multi-messenger, that means combining different "messengers" such as light, GW and neutrino, detections were to be prioritized in the next years. Currently operating flagship gamma-ray observatories have long passed their main operation lifetimes, and new replacements are still in very early phases.

Professionally or student developed CubeSats with miniaturized detector technology have proven to be able to successfully detect gamma-ray bursts (GRBs) [2–4]. However, gamma-ray photons cannot be focused as with a traditional optical telescope. A good way around to accurately observing and locating GRBs is to perform triangulation with multiple satellites in orbit. Therefore, having multiple missions with large fields of view and good resolution capabilities is of very much importance for the proper study of this astrophysical phenomena.

Missing clear explanation of what are gamma-ray bursts and why we need to send detectors to orbit?

II. MULTI-MESSENGER ASTRONOMY

III. COSMOLOGY: GRBS AS HIGH-z PROBES

A. Cosmological Models and Tensions

Since the late 1920s, we know that the universe is in expansion [5, 6], this lead to the development of the 'Big Bang theory' later confirmed by the observation of the Cosmic Microwave background (CMB) [7, 8]. In the 1990s, the expansion of the universe was found to be accelerated using Ia SNe observations [9, 10], motivating the description of the expansion rate as a time-dependant parameter $H(t)$ (Hubble parameter), with $H_0 = H(t_0)$. The relation between the distance d_L and the redshift $z \propto v_{\text{radial}}$ is then given by the luminosity distance function:

$$d_L(z, \overline{\Omega}) = (1+z)\frac{c}{H_0}\int_0^{z'} \frac{dz'}{E(z)}$$

where H_0 is the Hubble-Lemaître constant (the present time rate of expansion) and $E(z)$ is the dimensionless Hubble parameter, which is different regarding the cosmological model one considers. The current standard model in cosmology ΛCDM assumes the dominance of cold dark matter (CDM) and a cosmological constant denoted Λ [**?**]. Deriving from the Friedmann equations [**?**] (which is a description of the universe treating its components as fluids) we obtain

$$E^2(z) = \left(\frac{H(t)}{H_0}\right)^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda$$

where a is the scale factor set to $a(t_0) = a_0 = 1$ and Ω_i are the energy density parameters for each component of the Universe's energy. According to ΛCDM these are the components contributing to the total energy density of the universe [**?**]:

$$\begin{cases} \text{radiation} : \Omega_r \\ \text{matter} : \Omega_m \\ \text{curvture (geometric)} : \Omega_k \\ \text{dark energy} : \Omega_\Lambda \end{cases}$$

Most of the time, a *flat* ΛCDM model is assummed $\Omega_\Lambda + \Omega_m = 1$ with $\Omega_k = \Omega_r = 0$ (due to flatness and negligible constribution of radiation). When not trying to constraint the cosmology, and assuming a standard cosmological $\Omega_\Lambda = 1 - \Omega_= h_0 = 0.7$ is also assumed, agreeing enough with estimated values for these parameters and $h_0 := H_0/100$.

In recent times, attention has been drawn to the so-called 'Hubble Tension', an open problem in cosmology in which measurements of H_0 don't agree between early-universe CMB and late-time local distance ladder observations [11–13]. More recently, in the era of 'precision cosmology' this tension, alongside the σ_8 [**?**] and Ω [**?** **?**] tensions have become even more obvious. These tensions alongside the fine-tuning problem of the cosmological constant Λ [**?**] and the growing evidence of dynamical dark energy (DDE) [**?**] contrasting previous results and assumptions has motivated the search of alternative methods to measure cosmological parameters and test proposed alternative models.

B. GRBs as Cosmological Probes

Gamma-ray bursts are amongs the most luminous astromical events in the Universe and are visible at large distances, making their potential use as high-z cosmological probes compelling. However, GRBs are not standard candles, and we rely on empirical relations between observables to compute our luminosity distance and thus test different cosmologies. Multiple potential relations to 'standardize' GRBs have been proposed [**?**].

When utilizing GRBs to probe the cosmological parameters, we encounter the circularity problem, which is that we need to assume a cosmology $\overline{\Omega}$ in order to compute the fit to the empirical relation, which we want to use to constraint the parameters. For example, the Amati relation relates the peak spectral energy (E_{peak}) and the isotropic equivalent energy (E_{iso}) and is written as:

$$E_{\text{peak}} = a \cdot E_{\text{iso}}^b$$

this can be used to contraint the cosmology since E_{iso} is computed from the observed fluence $\mathcal{F}$ relying on the luminosity distance by

$$E_{\text{iso}}(\overline{\Omega}) = 4\pi \mathcal{F} d_L(z, \overline{\Omega})/(1+z)$$

In this case, in order to compute (a, b) we need to assume $\overline{\Omega}$. The circularity problem is avoided by computing the empirical relation fit for a set of possible cosmologies, among which the 'best cosmology' is assumed to be the one with the least scatter in the empirical relation.

Key instrumentation requirements for this purpose is a broad energy range, including high-energy γ-ray sensitivity and enough energy resolution, allowing to accurately measuring E_{peak}. Additionally, a broad field of view contributes to increasing the chances of detecting this short-lived transient events, more data . According to estimates [**?**], 0000 detections with average uncertainty of $\sigma_E = 0.00$ are required to constrain cosmological parameters to a 10% error with 99% certainty.

While GRBs are sensitive to dark energy and matter energy density parameters fairly well, since there are no close anchors, GRBs do badly constraint H_0, indeed it's introduction as a free parameter brings in a degeneracy with the empirical relation. That is why GRBs are better used as a complementary information to another probe, such as GW luminosity distance data. [**?**]

how much energy resolution for $X\%$ error on energy? What about other empirical relation's requirements?

C. Multi-Messenger Cosmology

what does GW work so well together with GRB when constraining cosmologies? first write II, since it build on that

IV. LORENTZ INVARIANCE AND HIGH-ENERGY PHOTON LAGS

A. A Theory of Quantum Gravity

- briefly explain why a theory of quantum gravity is of such relevance

B. Detecting Lorentz Invariance with High-Energy Photons

- elaborate on how detecting high-energy photon logs provides proof of a granular (quantuized) space-time
- transform the theory into instrumentation requirements (spectral range and resolution, timing resolution)

V. SUMMARY OF TECHNICAL REQUIREMENTS

Section: COSMOLOGY: GRBS AS HIGH-z PROBES

- high enough energy and spectral resolution to accurately identify E_{peak}
- broad energy range (especially towards higher energies) to don't be blind to E_{peak}
- broad field of view (to observe more GRBs)

VI. CONCLUSIONS

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